

Usama Muhammad

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Employment History

Aug 2017 – Aug 2026

📌 **Graduate Student Researcher**, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea.

- Developed advanced **Machine Learning** techniques for signal integrity and equalizer optimization (Samsung DS collaboration), applying latent representation learning for **Anomaly Detection** in DRAM signals.
- Led intelligent robotics research, developing robust visual odometry and localization methods for highly maneuverable robots utilizing **Deep Learning**.
- Built and deployed an end-to-end **ML** application for regulatory intelligence, extracting and summarizing safety and health policies from diverse government documents.
- Applied **Deep Reinforcement Learning** algorithms (**PPO**, **DDPG**, **Q-Learning**) to control systems including quadcopters and autonomous land robots.
- Engineered fair data representation methods to mitigate bias in downstream utility tasks, advancing transparency in **ML** models.

May 2026 – Present

📌 **AI Engineer**, Agent Astro (agentastro.ai), Remote (US).

- Designed the end-to-end **AI** and data architecture for an **FDA regulatory-intelligence** platform serving MedTech teams, spanning ingestion, retrieval, reasoning, and structured-output generation over FDA 510(k), PMA, recall, and guidance corpora.
- Built production **RAG** combining **pgvector** semantic retrieval, **Elasticsearch** full-text search, and **Neo4j** predicate-graph traversal, with per-claim provenance using **Google Gemini**, **Vertex AI**, and **OpenAI** APIs.
- Engineered a **multi-agent Deep Research** orchestrator (planner, analyst, verifier, reflector, synthesizer) with cited research-memo generation, integrating **Tavily** for web research. Published an **MCP (Model Context Protocol) server** giving partner companies and tools like **ChatGPT**, **Perplexity**, and **Claude** direct access to these capabilities.
- Built the **Device Twin** similarity/scoring service and **Predicate Investigation** pipeline (lineage tracing, substantial-equivalence analysis, scenario compare) on a Neo4j knowledge graph, plus an **eSTAR** FDA-submission builder with RTA pre-flight checks.
- Built high-throughput **ingestion pipelines** for openFDA (510(k), PMA, recalls, adverse events, UDI, enforcement) and GCS-hosted CDRH PDFs, with LLM-based document intelligence (**PyMuPDF** + Gemini extraction) and automated schema self-healing.
- Shipped async production services in **FastAPI** (Python 3.12, SQLAlchemy 2, Pydantic v2) backed by **PostgreSQL** + **pgvector**, **Neo4j**, and **Elasticsearch**. Deployed the full platform on **Google Cloud Platform (GCP)** with Docker, running **Cloud Scheduler** cron jobs for automated data refresh and database health checks.

Employment History (continued)

Oct 2025 – Present

📌 **AI Initiative Team Lead**, Braindeck Inc., Seocho-gu, Seoul, South Korea.

- Built a custom **ASR** model from scratch for Korean-language users with dysarthria, where no off-the-shelf solution was adequate, and integrated it into a real-time avatar communication platform with a **RAG**-based personalized communication tool.
- Fine-tuned and deployed **OpenAI Whisper** on custom datasets for domain-specific vocabulary and accent adaptation, and built **Voice Cloning** pipelines using **Fish Audio** (ASR, TTS, cloning) for on-premise deployment on limited GPU resources.
- Built an end-to-end pipeline for medically certified diagnosis, rehabilitation, and communication assistance for Korean-language users with dysarthria.
- Engineered the entire backend infrastructure for the avatar and medically certified rehabilitation applications utilizing **FastAPI**, and deployed these comprehensive **AI** services across **AWS**, **GCP**, and **Microsoft Azure**.
- Designed an advanced Multi-sensory Emotion Recognition Solution utilizing multisensor fusion (speech, facial expressions, biosignals) for precise emotion prediction.
- Architected a two-stage cascade **ML** system (**LightGBM** + **Random Forest**) for acoustic **NDT** defect classification of industrial fasteners (KPF), replacing a rule-based quality-control system with a data-driven **smart-manufacturing** pipeline over frequency-response (FRF) spectra.

Jul 2025 – Present

📌 **AI Lead**, Brain Box Automations, Pakistan (Remote).

- Managing multiple projects in the AI department, supervising developers and maintaining client relations.
- Implemented **CI/CD for ML** pipelines and extensively developed production-grade AI agent-based services using **FastAPI**, **LangGraph**, and **LangChain** with focus on **LLMOps** and **MLOps**.
- Deployed scalable AI solutions on **AWS**, **GCP**, and **Microsoft Azure** platforms using **Docker** and **Microservices**.
- Built multi-agent research systems, niche coaching and cognitive assistants, AI-powered content management, and end-to-end recruitment automation pipelines with custom AI models, MCPs, and LLM tools, extensively leveraging **Gemini ADK** (Agent Development Kit) for complex multi-agent orchestration.

Oct 2025 – Present

📌 **AI Researcher**, Tianjin Medical University Eye Hospital, Tianjin, China.

- Developing advanced **AI** algorithms for the detection and prognosis of eye diseases, including **Myopia** and **Diabetic Retinopathy**.
- Engineered a Spatio-Temporal Heterogeneous Data Fusion framework and a Time-Aware Transformer for predicting longitudinal pediatric myopia progression, achieving a 45.6% error reduction.
- Validated a cross-domain transfer learning approach utilizing **EfficientNet-B4** to generalize **Diabetic Retinopathy** screening from desktop to handheld fundus cameras, achieving 92.3% sensitivity and 94.2% specificity.
- Developed and validated a **Machine Learning**-based Six-Item Ocular Surface Disease Index (OSDI-6) severity grading scale using decision trees.

Employment History (continued)

- Aug 2023 – Sep 2025  **Freelance AI/ML Developer & Consultant**, Self-Employed (Remote).
- Developed Full-stack AI applications and multimodal platforms utilizing **React**, **Next.js**, and **GraphQL**.
 - Engineered advanced AI agents with **OpenAI API (GPT-4o)**, **Anthropic Claude API**, **Gemini API**, and open-source models (**Qwen**, **Mistral**) utilizing **RAG**, **Vector Databases** (**Qdrant**, **Pinecone**), and advanced **Context Engineering**.
 - Built production **Voice AI** pipelines for KindredPM (US) using **Deepgram** real-time WebSocket **STT** with voice activity detection and **Aura TTS** for low-latency conversational agents, and integrated **Gemini Live API** for real-time speech-to-speech with barge-in support and multilingual dialog.
 - Provided client consultation for **LLMOps** operations and production deployment.
- Feb 2016 – Jul 2017  **Research Assistant**, NUST - Robotics & Intelligent Systems Engineering (RISE Lab), Islamabad, Pakistan.
- Developed wearable fNIRS device for real-time hemodynamic brain profiling.
 - Implemented signal processing and ML techniques for brain signal feature extraction supporting motor neuron disability research.
- Jul 2016 – Sep 2016  **Intern Application Engineer**, National Instruments (NI), Islamabad, Pakistan.
- Completed LabVIEW certification training.
 - Designed and implemented “Hybrid Control of Inverted Pendulum” project on NI hardware platforms.

Leadership, Entrepreneurship & Service

- May 2026 – Present  **Founder / Tech Track Participant**, “Echo Project,” Everyone’s Startup Project (모두의 창업 프로젝트), Ministry of SMEs and Startups (Daejeon CCEI). Developing an **AI-driven** cognitive preservation platform; advanced via Fast-Track selection and secured **2,000,000 KRW** startup activity fund.
- Aug 2025 – Nov 2025  **Representative & Lead Developer**, Team ‘Viro,’ 2025 KAIST Startup Crew (Laboratory-Specialized Startup Leading University Program), KAIST. Directed technical architecture and application development; secured **2,000,000 KRW** in startup support funding.
- Aug 2023 – Feb 2024  **Part-time Researcher**, KAIST (Part-time Scholarship). Executed targeted research tasks under the supervision of Prof. Sang-Gug Lee and Prof. Thomas Cho.
- 2018, 2022  **KAIST EE Ambassador**, EEIO Ambassador Program. Represented KAIST EE at the National University of Sciences and Technology (NUST), Pakistan, conducting multi-campus recruitment seminars for prospective graduate students.
- 2021, 2024  **Short-term Teaching Assistant**, KAIST EE Visit Camp. Coordinated camp logistics and facilitated research lab tours for prospective international students.
- Sep 2018 – Feb 2019  **Teaching Assistant**, KAIST EEIO. Managed information desk operations, supported international students administratively, and organized EEIO Visit Camp events.

Education

- Aug 2019 – Aug 2026  **Ph.D., Electrical Engineering**, Korea Advanced Institute of Science and Technology (KAIST).
GPA: 3.95/4.3 (Top 5%)
Thesis: *Learning Latent Representations for Anomaly Detection and Reinforcement Learning-Based Equalizer Optimization in High-Speed DRAM Systems*
Advisor: Prof. Dong Eui Chang
- Aug 2017 – Aug 2019  **M.S., Electrical Engineering**, Korea Advanced Institute of Science and Technology (KAIST).
GPA: 3.97/4.3 (Top 3.7%)
Thesis: *Towards Robust Neural Networks and Efficient Exploration in Reinforcement Learning*
Advisor: Prof. Dong Eui Chang
- Sep 2013 – Jun 2017  **B.E., Electrical Engineering**, National University of Sciences and Technology (NUST), Pakistan.
GPA: 3.99/4.0 (**Gold Medalist**)
Thesis: *Development of Functional Near Infrared Spectroscopy (fNIRS) based System for Detection of Distinctive Brain Signal Patterns*
Advisor: Prof. Fahad Mumtaz Malik
Co-Advisor: Prof. Yasar Ayaz
Awarded President of Pakistan Gold Medal for Best Graduate in Electrical Engineering

Research Publications

Patents

- 1 **M. Usama** and D. E. Chang, "Reinforcement learning-based equalizer parameter optimization method and system (original title: 강화학습 기반 이퀄라이저 파라미터 최적화 방법 및 시스템)," Korea Patent Application 10-2025-0119661, Aug. 2025.
- 2 **M. Usama** and Y. R. Jung, "Method and apparatus for monitoring health status of patient based on reliability using multiple biosignals (original title: 복수의 생체 신호를 이용하여 신뢰도를 기반으로 환자의 건강 상태를 모니터링하는 방법 및 장치)," Korea Patent Application 10-2025-0213058, Dec. 2025.

Journal Articles (Accepted / Published)

- 1 E. E. Pazo, X. Liu, S. Liu, Q. Zhang, M. Zhang, Z. Zhang, S. Moutari, **M. Usama**, and X. Ren, "Validation of the Eyerobo FC portable fundus camera for diabetic retinopathy screening using public datasets and deep learning," *Ophthalmology and Therapy*, 2026.  DOI: 10.1007/s40123-026-01353-w.
- 2 H. U. K. Shinwari and **M. Usama**, "Memory-augmented architecture for long-term context handling in large language models," *IEEE Transactions on Artificial Intelligence*, 2026, Equal contribution.  DOI: 10.1109/TAI.2026.3701680.
- 3 **M. Usama** and D. E. Chang, "Distributional reinforcement learning with information bottleneck for uncertainty-aware DRAM equalization," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 16, no. 8, pp. 1912–1929, 2026.  DOI: 10.1109/TCPMT.2026.3674506.
- 4 **M. Usama** and D. E. Chang, "Multi-lane joint DRAM equalization via cooperative multi-agent distributional reinforcement learning," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, 2026.  DOI: 10.1109/TCPMT.2026.3711630.
- 5 N. Wei, C. Li, S. Moutari, **M. Usama**, J. Li, Q. Chen, E. E. Pazo, and X. Qian, "Predicting pediatric myopia progression from single screening encounters: A multi-outcome development and validation study," *Journal of Translational Medicine*, 2026.  DOI: 10.1186/s12967-026-09003-2.

- 6 M. Xia, B. Yao, Y. Wang, F. Liang, W. Zhao, L. Qin, S. Liu, **M. Usama**, Z. Zhang, W. Guo, E. E. Pazo, and X. Ren, "Machine learning-based prediction of cycloplegic refraction using the Eyerobo vision screener: Design—a cross-sectional comparative device study," *Ophthalmology and Therapy*, 2026.
DOI: 10.1007/s40123-026-01482-2.
- 7 E. E. Pazo, S. Moutari, F. Gao, H. Liying, **M. Usama**, X. Li, and J. Liu, "Structure-preserving super-resolution of retinal fundus images via a dual-transformer residual network," *Frontiers in Medicine (Section: Ophthalmology)*, 2025. DOI: 10.3389/fmed.2025.1730678.
- 8 H. U. K. Shinwari and **M. Usama**, "ARD-LoRA: Dynamic rank allocation for parameter-efficient fine-tuning of foundation models with heterogeneous adaptation needs," *IEEE Transactions on Artificial Intelligence*, 2025, Equal contribution. DOI: 10.1109/TAI.2025.3605569.
- 9 **M. Usama**, H.-D. Jang, and D. E. Chang, "Deep reinforcement learning based DFE structure optimization for high-speed DRAM signals," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, 2025. DOI: 10.1109/TCPMT.2025.3648009.
- 10 **M. Usama** and D. E. Chang, "Learning flexible and fair data representations," *IEEE Access*, vol. 10, 2022.
DOI: 10.1109/ACCESS.2022.3207747.

Journal Articles (Under Review)

- 1 **M. Usama**, "Delta-CKA and the warm-start imperative: Empirical findings on layer compression in large language models," *Transactions on Machine Learning Research (TMLR)*, 2026, Under Review.
- 2 **M. Usama**, "The GRPO tax is smaller than you think: A longitudinal study of capability preservation during reasoning training," *Transactions on Machine Learning Research (TMLR)*, 2026, Under Review.
- 3 **M. Usama** and D. E. Chang, "Convergence without understanding: When language models agree on representations but disagree on reasoning," *Journal of Machine Learning Research (JMLR)*, 2026, Under Review.
- 4 **M. Usama** and D. E. Chang, "The system prompt illusion: How instruction preambles modify computation in language models," *Journal of Machine Learning Research (JMLR)*, 2026, Under Review.
- 5 **M. Usama** and D. E. Chang, "Deep reinforcement learning-based DRAM equalizer parameter optimization using latent representations," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2025, Under Review.
- 6 **M. Usama**, H.-D. Jang, S. Shanbhag, Y.-C. Sung, S.-J. Bae, and D. E. Chang, "Learning high-quality latent representations for anomaly detection and signal integrity enhancement in high-speed signals," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2025, Under Review.
- 7 **M. Usama** and Y. Jung, "Evidential multimodal transformers with dissonance-aware belief fusion for emotion recognition," *IEEE Transactions on Affective Computing*, 2025, Under Review.

Conference Proceedings

- 1 **M. Usama** and S. Y. Jung, "Beyond adjacent layers: Graph-guided layer fusion for compressing large language models," in *Conference on Neural Information Processing Systems (NeurIPS)*, Accepted, 2026.
URL: <https://github.com/Usama1002/fcm-layer-fusion>.
- 2 **M. Usama** and Y. Jung, "F-convnext: Frequency-aware domain generalization for deepfake detection via self-blended synthetic training," in *Conference on Neural Information Processing Systems (NeurIPS)*, Under Review, 2026.
- 3 **M. Usama** and D. E. Chang, "Uncertainty-aware DRAM equalizer optimization," in *2025 Summer Annual Conference of the Korean Institute of Electromagnetic Engineering and Science (KIEES)*, 2025.
- 4 **M. Usama** and D. E. Chang, "Uncertainty-aware DRAM equalizer optimization using bayesian representation learning and deployment risk assessment," in *2025 Conference on Electronics, Semiconductors, and Artificial Intelligence (CESAI)*, 2025.

- 5 **M. Usama** and D. E. Chang, "Optimization of decision feedback equalizer structure through genetic algorithm and neural network evaluation," in *39th Institute of Control, Robotics and Systems (ICROS) Conference 2024*, 2024.
- 6 **M. Usama** and D. E. Chang, "Particle swarm optimization for DFE structure optimization with neural fitness evaluation," in *2024 Summer Annual Conference of IEIE*, 2024.
- 7 **M. Usama** and D. E. Chang, "Representation learning for distortion classification in DRAM data," in *39th Institute of Control, Robotics and Systems (ICROS) Conference 2024*, 2024.
- 8 **M. Usama** and D. E. Chang, "PyEye: An integrated approach for signal integrity assessment and eye diagram generation," in *2023 IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia)*, 2023.
- 9 **M. Usama** and D. E. Chang, "Learning-based control of quadrotor using gaussian processes," in *6th Joint Conference of Korean Artificial Intelligence Association (CKAIA 2022)*, Recipient of Special Society Award, 2022.
- 10 **M. Usama** and D. E. Chang, "Learning-driven exploration for reinforcement learning," in *2021 21st International Conference on Control, Automation and Systems (ICCAS)*, 2021.
- 11 D. Lee, **M. Usama**, and D. E. Chang, "Control of furuta pendulum with reinforcement learning," in *19th International Conference on Control, Automation, and Systems (ICROS 2019)*, 2019.
- 12 **M. Usama** and D. E. Chang, "Robotic navigation using entropy-based exploration," in *2019 19th International Conference on Control, Automation and Systems (ICCAS)*, 2019.
- 13 **M. Usama** and D. E. Chang, "Towards robust neural networks with lipschitz continuity," in *17th International Workshop on Digital Forensics and Watermarking (IWDW 2018)*, 2018.

Skills

AI/ML	 Deep Learning, Reinforcement Learning, Large Language Models (LLMs), Multi-Agent Systems (LangGraph , Cognitive Architectures), Foundation Model Fine-tuning, Parameter-Efficient Training (LoRA, ARD-LoRA), Computer Vision, NLP, HuggingFace Transformers , Qwen , Mistral , LightGBM , Gradient-Boosted Trees, PPO , DDPG .
LLM Ops	 Context Engineering (ReAct Context Engineering), RAG Systems, Vector Databases, MCP Server Development (published integrations for ChatGPT, Claude, Perplexity), Custom LLM Tools, OpenAI API , Anthropic Claude API , Gemini API , Gemini ADK , Production LLM Deployment, LLMOps , MLOps , NLzSQL .
Voice AI	 Deepgram (Real-time STT/TTS), OpenAI Whisper (Fine-tuning), Fish Audio (ASR/TTS/Voice Cloning), Gemini Live API (Speech-to-Speech), Custom ASR Training, Voice Activity Detection, Low-latency Conversational AI.
Coding	 Python, PyTorch, TensorFlow, TypeScript, JavaScript, React, Next.js , Node.js, FastAPI, Flask, Django , GraphQL , LaTeX .
Tools	 Git, Docker , CI/CD , Linux, LabVIEW, MATLAB, Gradio, Streamlit, Jupyter, VS Code.
Cloud & DB	 AWS , GCP (Vertex AI, Cloud Scheduler), Azure , PostgreSQL, pgvector , Neo4j , Elasticsearch , SQLite, MongoDB, Qdrant (Vector DB), Pinecone , Redis.
Research	 Academic writing, peer-reviewed publication, experimental design, statistical analysis, IEEE and journal submissions.
Languages	 English (Fluent), Urdu (Native), Korean (Intermediate – TOPIK Level 2).

Honors & Awards

Awards and Achievements

- 2026  **Gold Reviewer**, ICML 2026 (International Conference on Machine Learning). Top reviewer recognition based on area chair ratings; awarded complimentary conference registration.
- 2025  **Winner – Logo Design Contest**, KAIST Space Institute. Worth 1,000,000 KRW.
-  **Excellent Idea Award**, AI Technology Utilization in KAIST Administration Idea Contest, KAIST Center for Administration Development. Worth 300,000 KRW.
- 2017  **President of Pakistan Gold Medal**, Best Graduate in Bachelor of Engineering, NUST.
- 2022  **Special Society Award**, Worth 1 million KRW, Korean Artificial Intelligence Association.
- 2019  **AlphaPilot Qualifier**, Lockheed Martin AI Drone Racing Innovation Challenge. Team Puffin ranked 7th among 400+ teams.
- 2017  **Runners Up**, All Pakistan Engineering Category, AirTECH 2016, Air University.
-  **Runners Up**, All Pakistan Embedded Systems Category, COMPPEC'17, CEME NUST.
- 2013–2016  **EME High Achievers Awards**, Consecutive recognition for academic excellence.
- 2013  **5th Position**, Among 102,080 students in FBISE HSSC Examination.

Certification

- 2016  **Certified Associate Developer – LabVIEW**. Awarded by National Instruments.
-  **PLC, HMI and SCADA Systems**. Professional Training.

References

Available on Request